

Advanced Image Generation Models

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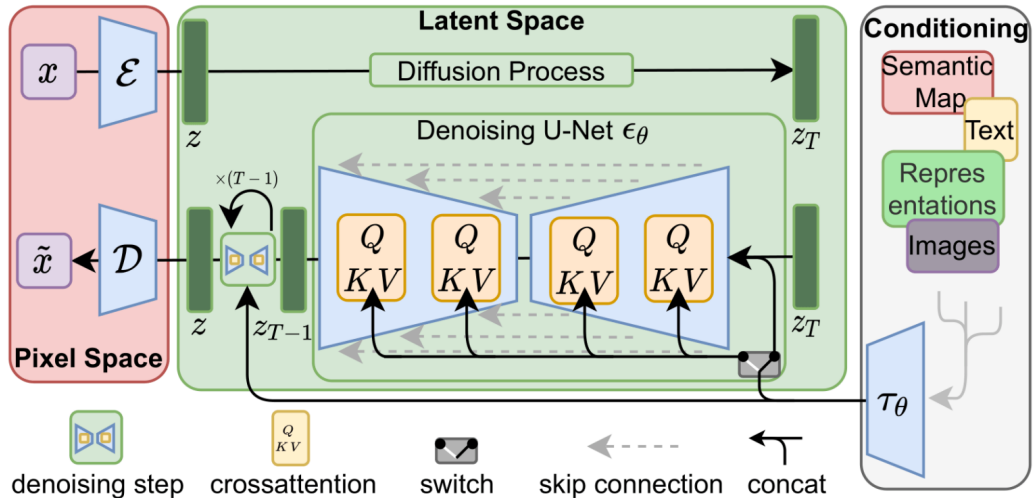
جامعة الملك عبد الله
للعلوم والتقنية

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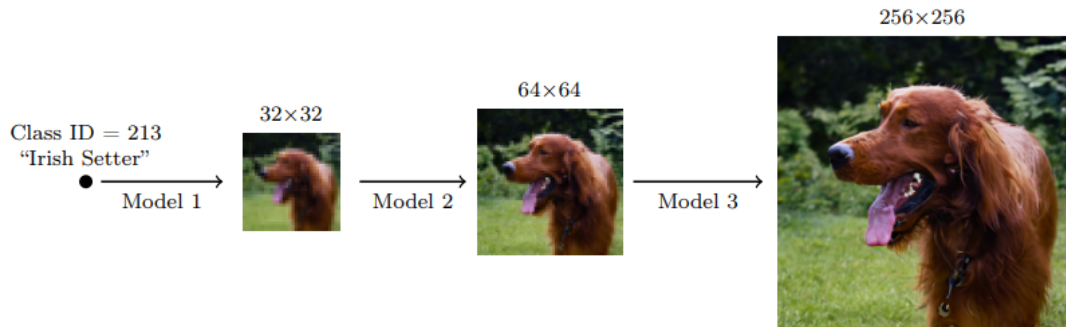
- ▶ The previous lectures covered why diffusion displaced GANs and how the core denoising recipe works; this lecture pushes beyond the base recipe.

Advanced Image Generation Models

- ▶ *Why it matters:*
 - Diffusion models are reshaping creativity, art, and design.
 - From AI avatars to realistic landscapes, they have endless applications.
 - You'll gain insight into how these models work — and how to use them yourself!

After this session, you will be able to:

- ▶ Describe cascaded diffusion pipelines for high-resolution generation.
- ▶ Summarize score-based diffusion models and their applications (DreamFusion, text-to-SVG, video adapters).
- ▶ Explain PDE-based approaches to diffusion modeling.
- ▶ Utilize text-conditioned diffusion models for controlled image generation.
- ▶ Explain CLIP-guided generation and editing with VQ-GAN and StyleCLIP.



$$p_{\theta}(x_0) = \int p_{\theta}(z_1)p_{\theta}(z_0 \mid z_1)p_{\theta}(x_0 \mid z_0)dz$$

Ho, Jonathan, et al. "Cascaded diffusion models for high fidelity image generation." The Journal of Machine Learning Research 23.1 (2022): 2249-2281.

- ▶ Train one unconditional model; the remaining models are low-resolution conditional super-resolution models.
- ▶ Condition on the low-resolution image, perform bilinear or bicubic upsampling, and concatenate the result to the noised input.
- ▶ Models can be trained independently.
- ▶ Hyperparameters can be tuned specifically for each resolution.

Conditioning Augmentation: Augment the low-res input z

► Method 1: Blurring

- Apply a Gaussian blurring filter
- Effective for models at 128, 256 resolution

► Method 2: Non-truncated Conditioning Augmentation

- Train the super-resolution $p(x|z_0)$ model conditioned on samples from the prior $p(z_0)$ with added corruption from the forward process $q(z_s|z_0)$
- S (corruption / noise strength) is a hyperparameter
- Effective for resolutions < 128

Cascaded Diffusion Models (cont.)



(a) 16×16 base



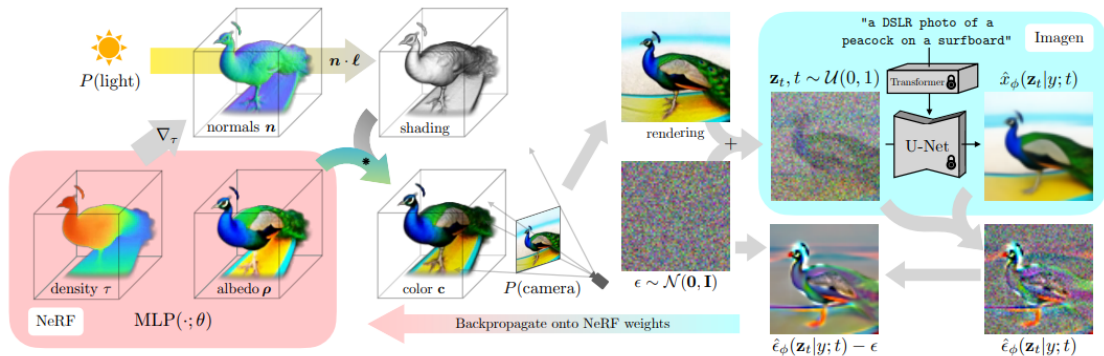
(b) $16 \times 16 \rightarrow 64 \times 64$ super-resolution, $s = 0$

Large improvements in using conditioning augmentation

Conditioning	FID vs train	FID vs validation	IS
No cascading	2.35	2.91	52.72 ± 1.15
16×16→64×64 cascading			
$s = 0$	6.02	5.84	35.59 ± 1.19
$s = 101$	3.41	3.67	44.72 ± 1.12
$s = 1001$	2.13	2.79	54.47 ± 1.05

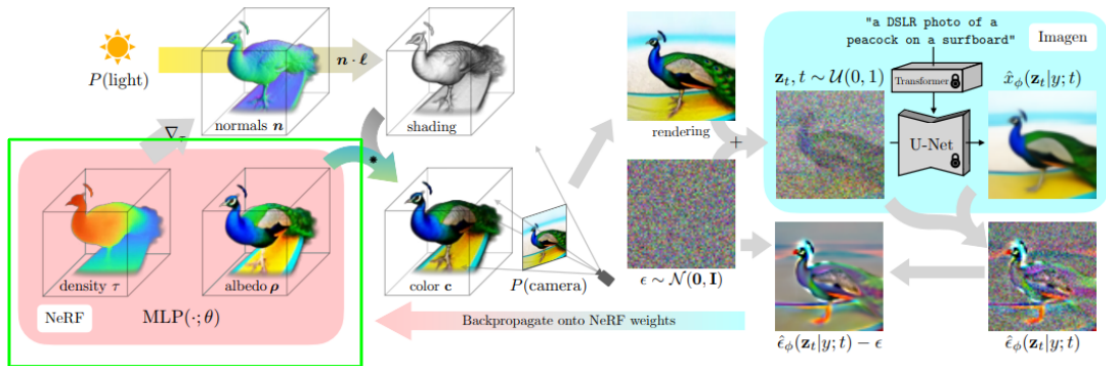
Advanced Diffusion Models: **Score-Based Diffusion Models**

How can you use a pretrained text-image model to generate 3D objects?

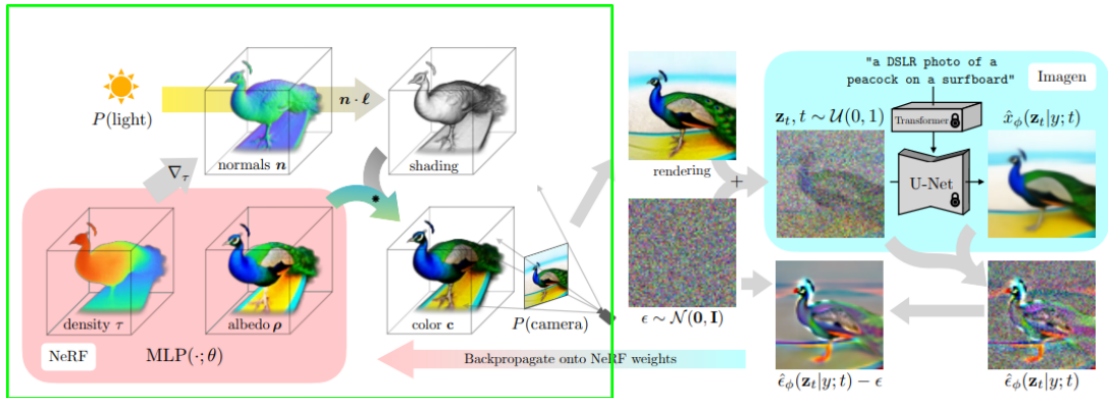


Poole, Ben, et al. "Dreamfusion: Text-to-3d using 2d diffusion." arXiv preprint arXiv:2209.14988 (2022).

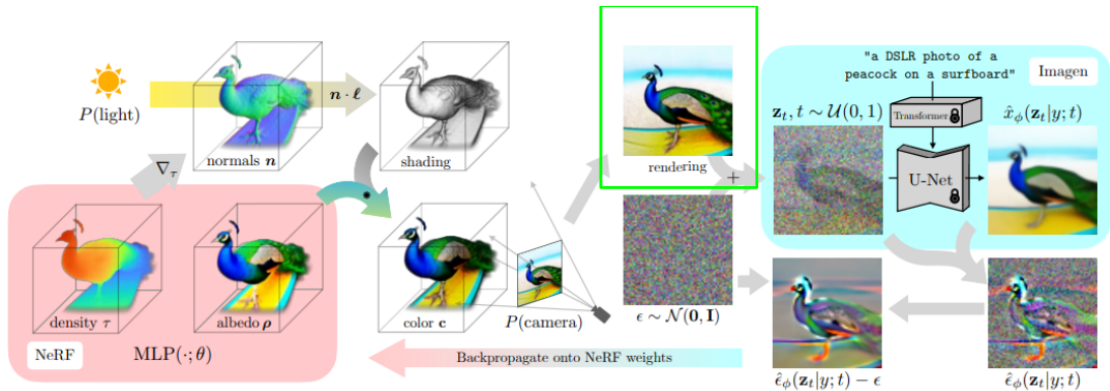
Start with a randomly initialized parameterized 3D representation (NeRF)



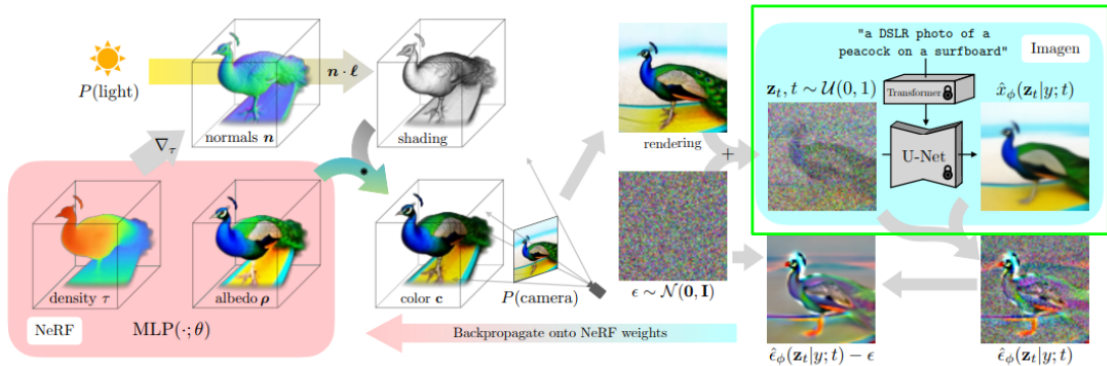
Randomly sample camera angle and lighting



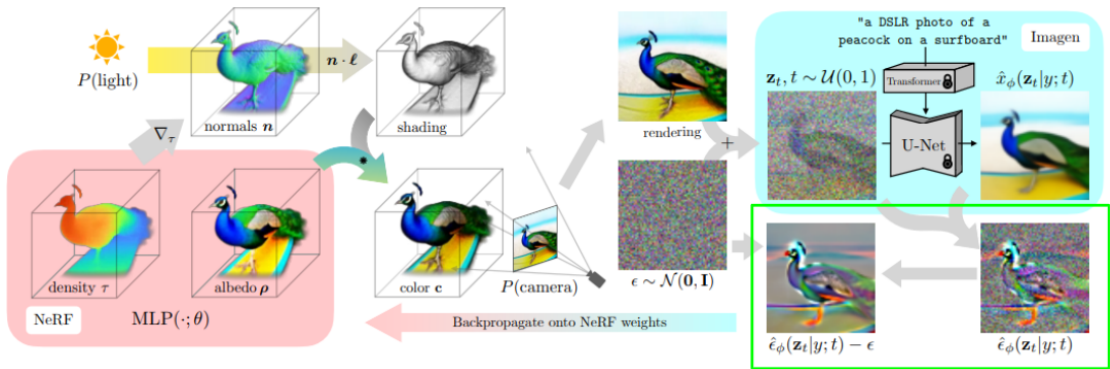
Render the image (64x64)



Noise the image and feed it through the diffusion loss to predict the noise



Compute the SDS loss and backpropagate



Intuitively, we want the NeRF parameters to produce images that minimize the diffusion loss

$$\mathcal{L}_{\text{Diff}}(\phi, \mathbf{x}) = \mathbb{E}_{t \sim \mathcal{U}(0,1), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [w(t) \|\epsilon_{\phi}(\alpha_t \mathbf{x} + \sigma_t \epsilon; t) - \epsilon\|_2^2]$$

$$\nabla_{\theta} \mathcal{L}_{\text{Diff}}(\phi, \mathbf{x} = g(\theta)) = \mathbb{E}_{t, \epsilon} \left[w(t) \underbrace{(\hat{\epsilon}_{\phi}(\mathbf{z}_t; y, t) - \epsilon)}_{\text{Noise Residual}} \underbrace{\frac{\partial \hat{\epsilon}_{\phi}(\mathbf{z}_t; y, t)}{\partial \mathbf{z}_t}}_{\text{U-Net Jacobian}} \underbrace{\frac{\partial \mathbf{x}}{\partial \theta}}_{\text{Generator Jacobian}} \right]$$

But the objective is generally brittle / expensive (backprop through the UNet)

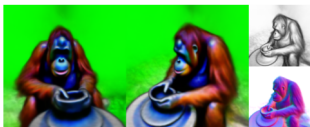
It works better to just remove the U-Net jacobian

$$\nabla_{\theta} \mathcal{L}_{\text{Diff}}(\phi, \mathbf{x} = g(\theta)) = \mathbb{E}_{t, \epsilon} \left[\underbrace{w(t) (\hat{\epsilon}_{\phi}(\mathbf{z}_t; y, t) - \epsilon)}_{\text{Noise Residual}} \underbrace{\frac{\partial \hat{\epsilon}_{\phi}(\mathbf{z}_t; y, t)}{\mathbf{z}_t}}_{\text{U-Net Jacobian}} \underbrace{\frac{\partial \mathbf{x}}{\partial \theta}}_{\text{Generator Jacobian}} \right]$$

$$\nabla_{\theta} \mathcal{L}_{\text{SDS}}(\phi, \mathbf{x} = g(\theta)) \triangleq \mathbb{E}_{t, \epsilon} \left[w(t) (\hat{\epsilon}_{\phi}(\mathbf{z}_t; y, t) - \epsilon) \frac{\partial \mathbf{x}}{\partial \theta} \right]$$

There is a theoretical motivation (see Appendix in [paper](#))

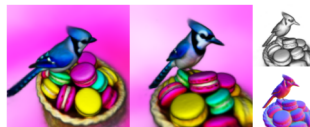
DreamFusion: Text-to-3D Generation (cont.)



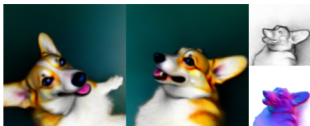
an orangutan making a clay bowl on a throwing wheel*



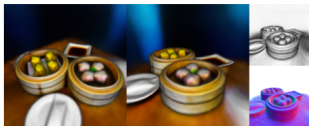
a raccoon astronaut holding his helmet†



a blue jay standing on a large basket of rainbow macarons*



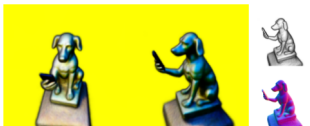
a corgi taking a selfie*



a table with dim sum on it†



a lion reading the newspaper*



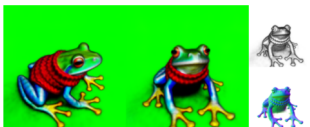
Michelangelo style statue of dog reading news on a cellphone



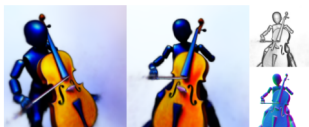
a tiger dressed as a doctor*



a steam engine train, high resolution*



a frog wearing a sweater*



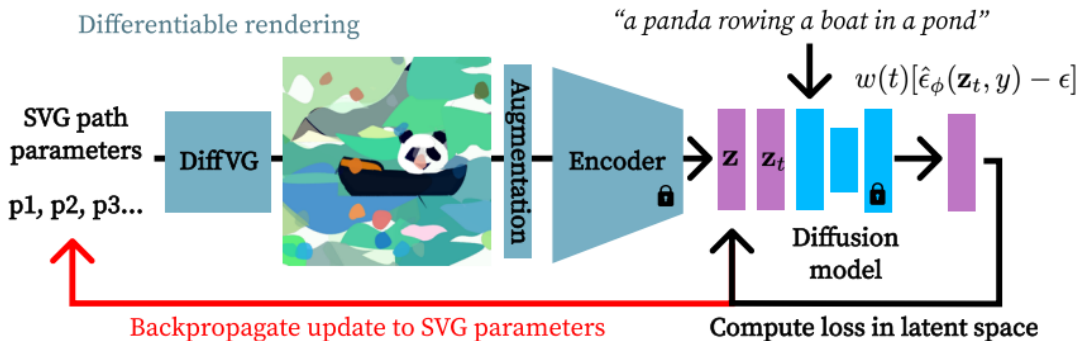
a humanoid robot playing the cello*



Sydney opera house, aerial view†

- ▶ Text-to-SVG generation is a novel application of diffusion models, enabling the creation of vector graphics from textual descriptions.
- ▶ The process involves training a diffusion model to learn the distribution of SVG images conditioned on text prompts.
- ▶ The model generates SVG paths, which are inherently vector-based and scalable without loss of quality.

- ▶ Using SDS loss is not limited to text-to-3D.
- ▶ SDS provides a general way to backpropagate and learn through any differentiable “rendering” pipeline.



Jain, Ajay, Amber Xie, and Pieter Abbeel. "Vectorfusion: Text-to-svg by abstracting pixel-based diffusion models." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2023.



a train*



an owl standing on a wire*



Underwater Submarine*



a boat*



A photo of a Ming Dynasty vase on a



A smiling sloth wearing a leather



a tuba with red flowers protruding from



a blue poison dart frog sitting on a wa-

DDPMs as Energy-Based Models (EBMs)

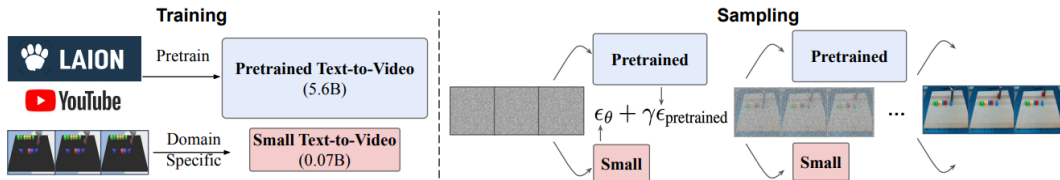
$$p_{\theta}(\tau) \propto e^{-E_{\theta}(\tau)}$$

Since the denoising function (ϵ) models the score, we have

$$\epsilon_{\theta}(\tau^t, t) = \nabla_{\tau} E_{\theta}(\tau^t)$$

Learn a distribution as a product of a pretrained (large) model and domain specific (small) model

$$\underbrace{p_{\text{product}}(\tau|\text{text})}_{\text{Product Distribution}} \propto \underbrace{p_{\text{pretrained}}(\tau|\text{text})}_{\text{Pretrained Prior}} \underbrace{p_{\theta}(\tau|\text{text})}_{\text{Video Model}}.$$



Under the product assumption, we can use the EBM interpretation to combine the scores of the diffusion models

$$p_{product}(\tau|\text{text}) \propto e^{-E_1(\tau)} e^{-E_2(\tau)} = e^{-(E_1(\tau)+E_2(\tau))}$$

$$\nabla_{\tau} E_{\theta}(\tau) = -(\nabla E_1(\tau) + \nabla E_2(\tau))$$

Algorithm 1 Sampling algorithm of Video Adapter

Input: Pretrained Text-to-Video Model $\epsilon_{\text{pretrained}}(\tau, t|\text{text})$, Inverse Temperature α , Prior strength γ .

Initialize sample $\tau_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

for $t = T, \dots, 1$ **do**

$\tilde{\epsilon}_{\text{text}} \leftarrow \epsilon_{\theta}(\tau_t, t|\text{text}) + \gamma \epsilon_{\text{pretrained}}(\tau_t, t|\text{text})$

// compute score using text-conditioned prior

$\epsilon \leftarrow \epsilon_{\theta}(\tau_t, t)$

// compute unconditional score

$\tilde{\epsilon}_{\text{cfg}} \leftarrow \epsilon + \alpha(\tilde{\epsilon}_{\text{text}} - \epsilon)$

// compute classifier guidance score

$\tau_{t-1} = \text{ddpm_sample}(\tau_t, \tilde{\epsilon}_{\text{cfg}})$

// run diffusion sampling (can use other samplers)

end for

Model	Bridge			Ego4D		
	FVD ↓	FID ↓	Param (B) ↓	FVD ↓	IS ↑	Param (B) ↓
Small (S)	186.8	38.8	0.07	228.3	2.28	0.07
Small (S) + Pretrained	177.4	37.6	0.07	156.3	2.82	0.07
Small (L)	152.5	30.1	0.14	65.1	3.31	2.8
Small (L) + Pretrained	148.1	29.5	0.14	52.5	3.53	2.8
Pretrained	350.1	42.6	5.6	91.7	3.12	5.6

Can adapt to different domains (robot trajectories, ego-centric camera)



[Link: GIFF](#)



[Link: GIFF](#)

Advanced Diffusion Models: **PDE-based Diffusion Models**

- ▶ Leverage diffusion models, originally for image generation, to solve and analyze PDEs.
- ▶ Learn complex PDE relationships by adding noise to solutions and reversing the process.
- ▶ Effective for handling incomplete or uncertain data.
- ▶ Valuable in scientific and engineering applications.

Core Idea

- ▶ Diffusion models transform data into a simple distribution (e.g., Gaussian noise) via a forward process.
- ▶ A reverse process is learned to reconstruct the original data from noise.
- ▶ For PDEs, the model maps input or boundary conditions to PDE solutions, capturing the equation's dynamics.

- ▶ **Generative Solutions:** Can produce multiple plausible PDE solutions, useful under uncertainty or incomplete information.
- ▶ **Uncertainty Modeling:** Capture a distribution of possible solutions, not just a single answer.
- ▶ **Partial Data Learning:** Train effectively on datasets with missing or incomplete data, suitable for real-world scenarios.
- ▶ **Flexibility:** Adaptable to both forward (solving PDEs) and inverse (parameter estimation) problems.

The diffusion equation is a fundamental PDE describing the distribution of a quantity (such as heat or particles) diffusing through a medium:

$$\frac{\partial u}{\partial t} = D \nabla^2 u \quad (1)$$

where:

- ▶ $u = u(\mathbf{x}, t)$ is the quantity of interest (e.g., concentration, temperature)
- ▶ D is the diffusion coefficient
- ▶ ∇^2 is the Laplacian operator

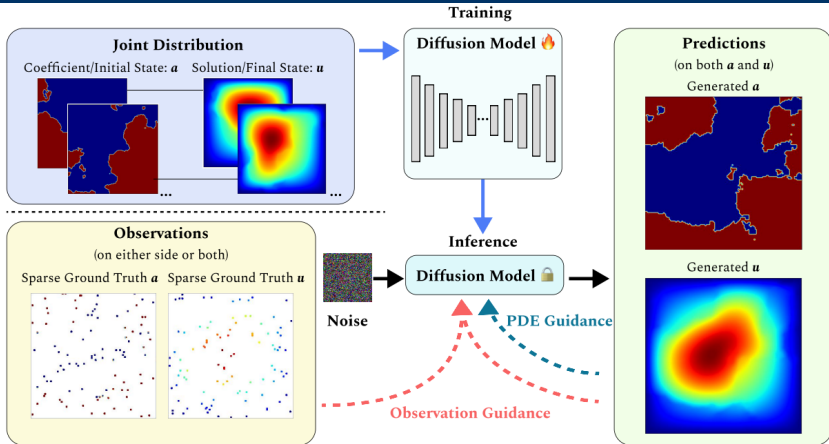


Figure 1: We propose DiffusionPDE, a generative PDE solver under partial observations. Given a family of PDE with coefficient (initial state) a and solution (final state) u , we train the diffusion model on the joint distribution of a and u . During inference, we gradually denoise a Gaussian noise, guided by sparse observation and known PDE function, to recover the full prediction of both a and u that align well with the sparse observations and the given equation.

Advanced Diffusion Models: **Text-Conditioned**

Text-Conditioned Diffusion Models

- ▶ **Overview:** Text-conditioned diffusion models enable the generation of highly specific images from textual prompts, such as *"a cat wearing sunglasses standing in Times Square"*. These models leverage powerful text encoders, such as CLIP or transformer-based architectures, to extract semantic representations from the input text.
- ▶ **Text Conditioning:**
 - The text encoder maps the prompt y to an embedding $e_y = \text{TextEncoder}(y)$.
 - This embedding is injected into the denoising UNet via cross-attention layers (the mechanism is detailed in the Stable Diffusion lecture).
- ▶ **The only change to the diffusion recipe:** everything from the Diffusion Models lecture carries over; the noise predictor simply takes the text embedding as an extra input, $\epsilon_\theta(x_t, t, e_y)$, in both the training objective and every denoising step.
- ▶ **Effect of Conditioning:** By conditioning the denoising process on e_y , the model generates images that faithfully reflect the textual description. The cross-attention mechanism in the UNet ensures that relevant parts of the text influence corresponding regions in the image, enabling complex and coherent scene synthesis.

Text-Conditioned Diffusion Models (cont.)

Advanced Image Generation:

CLIP-Guided Generation and Editing

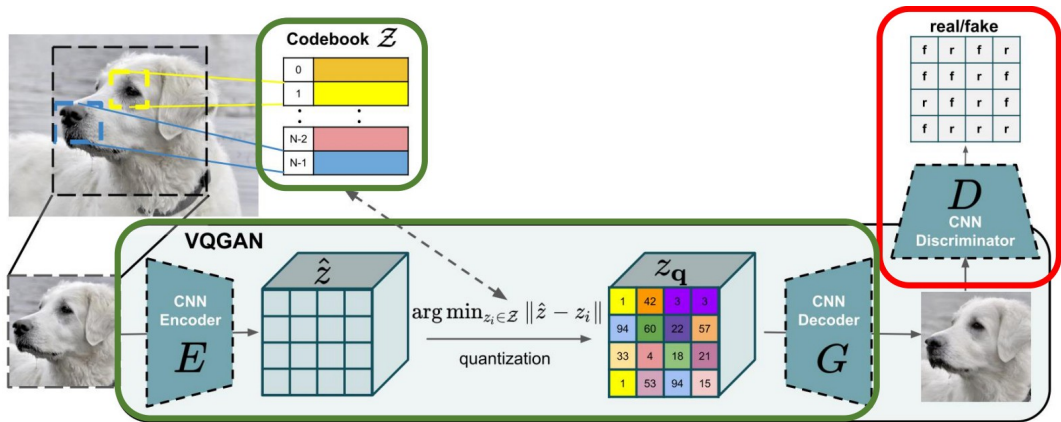
VQ-GAN – Vector Quantized Generative Adversarial Network

VQ-GAN is a generative model that combines the strengths of vector quantization and GANs to generate high-quality images.

- ▶ **Vector Quantization:** Encodes images into discrete latent codes, allowing for efficient representation and generation.
- ▶ **GAN Framework:** Uses a discriminator to guide the generator in producing realistic images.
- ▶ **Applications:** Image synthesis, super-resolution, and style transfer.

Architecture: VQ-GAN consists of an encoder, a quantizer, a decoder, and a discriminator.

- ▶ **Encoder:** Maps input images to a continuous latent space.
- ▶ **Quantizer:** Discretizes the continuous latent space into a finite set of codes.
- ▶ **Decoder:** Reconstructs images from the quantized latent codes.
- ▶ **Discriminator:** Evaluates the realism of generated images, providing feedback to the generator.



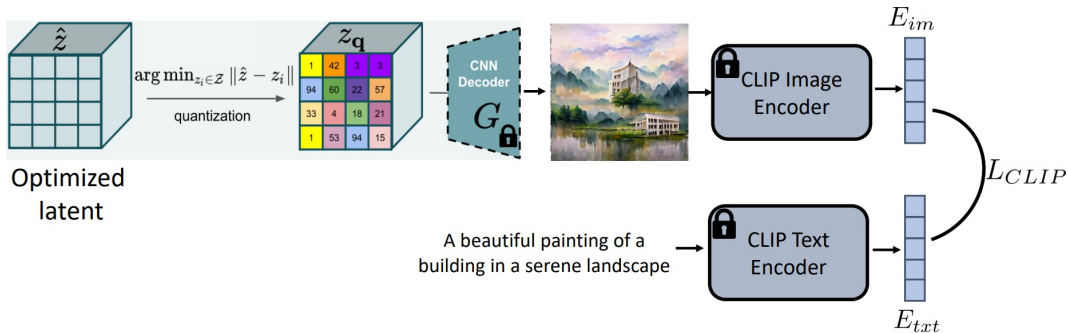
VQ-VAE + GAN

VQ-GAN + CLIP is a powerful combination of two models that enhances image generation and understanding by leveraging the strengths of both vector quantization and contrastive learning.

- ▶ **VQ-GAN:** A generative model that uses vector quantization to encode images into discrete latent codes, enabling efficient representation and high-quality image synthesis.
- ▶ **CLIP:** A model that learns visual representations by aligning images and text, allowing for zero-shot classification and improved understanding of visual content.

How They Work Together:

- ▶ **Image Generation:** VQ-GAN generates images by decoding quantized latent codes, while CLIP provides a text-based conditioning mechanism that guides the generation process.
- ▶ **Text Conditioning:** CLIP's text embeddings can be used to condition the VQ-GAN generator, allowing it to produce images that match specific textual descriptions.
- ▶ **Loss Function:** The similarity between generated image embeddings and text embeddings from CLIP can be used as a loss function to optimize the VQ-GAN generator, ensuring that the generated images align with the given text prompts.



$$L_{CLIP} = 1 - \text{Cos}(E_{im}, E_{txt})$$

[VQGAN-CLIP: Open Domain Image Generation and Editing with Natural Language Guidance, Crowson et al. ICCV 13 2021]

Applications:

- ▶ **Text-to-Image Synthesis:** VQ-GAN + CLIP can generate images based on textual descriptions, enabling creative applications like art generation and design.
- ▶ **Image Editing:** The combination allows for editing images based on text prompts, such as changing the style or content of an image while preserving its structure.
- ▶ **Zero-Shot Learning:** CLIP's zero-shot capabilities can be leveraged to classify generated images into categories without additional training, enhancing the versatility of the model.



"A sketch of 3D printer by da Vinci"



"An autogyro flying, artstation"



"A futuristic city in synthwave style"



"A colored pencil drawing of a waterfall"



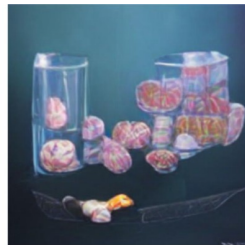
"A painting of a city in a deep valley"



"Baba Yaga's house, fantasy art"

Generation

A beautiful
painting of a
building in a
serene landscape



Editing



"A cake made of ice"



VQ-GAN + CLIP Editing

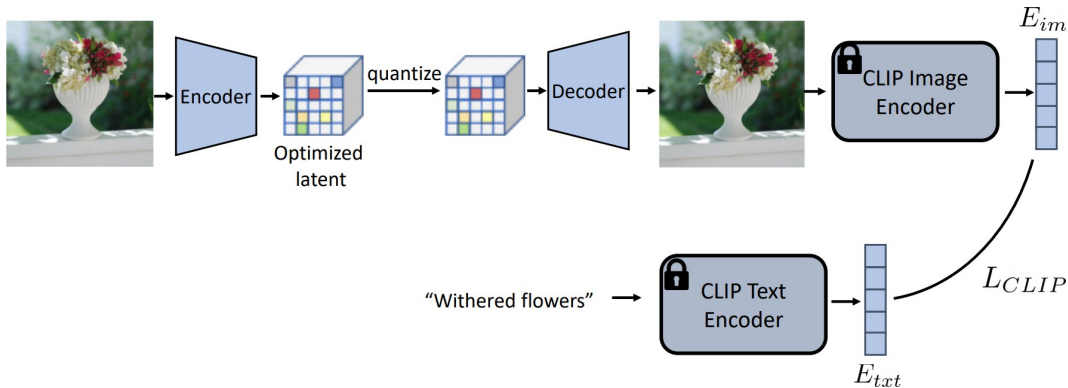
VQ-GAN + CLIP Editing combines the strengths of VQ-GAN for image generation and CLIP for text-guided editing, enabling high-quality image synthesis and manipulation.

- ▶ **VQ-GAN:** Generates images from discrete latent codes, allowing for efficient representation.
- ▶ **CLIP:** Guides the editing process by understanding text prompts, enabling targeted modifications to generated images.
- ▶ **Applications:** Image synthesis, style transfer, and content-aware editing.

How They Work Together:

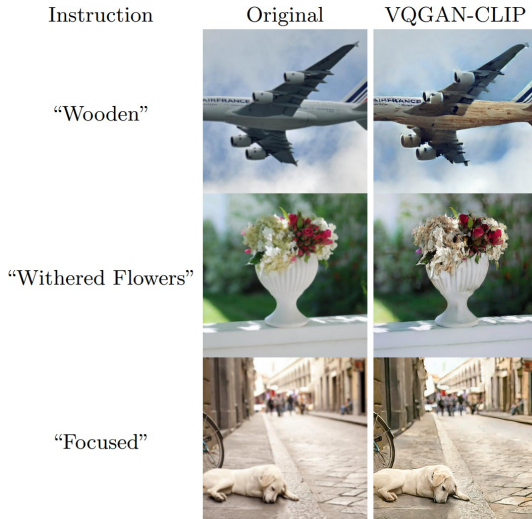
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- ▶ **Loss Function:** The similarity between generated image embeddings and text embeddings from CLIP is used as a loss function to optimize the VQ-GAN generator, ensuring that the generated images align with the given text prompts.

VQ-GAN + CLIP Editing (cont.)



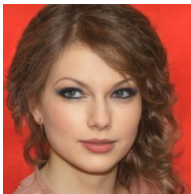
[VQGAN-CLIP: Open Domain Image Generation and Editing with Natural Language Guidance, Crowson et al. ICCV 13 2021]

VQ-GAN + CLIP Editing (cont.)

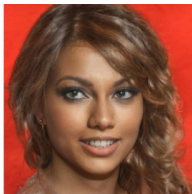


StyleCLIP is a method that combines the power of CLIP (Contrastive Language-Image Pretraining) with StyleGAN to generate images based on text prompts, allowing for fine-grained control over image attributes.

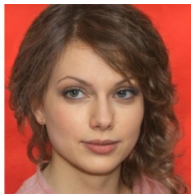
- ▶ **CLIP Embeddings:** Uses CLIP to encode text prompts into embeddings that capture semantic information.
- ▶ **StyleGAN Manipulation:** Applies these embeddings to manipulate the latent space of StyleGAN, enabling the generation of images that match the desired attributes described in the text.
- ▶ **Applications:** Useful for tasks like image editing, attribute manipulation, and creative content generation.
- ▶ **Benefits:** Provides a powerful way to generate diverse images based on textual descriptions.
- ▶ **Costs:** Requires careful tuning of the StyleGAN latent space and may produce artifacts if not managed properly.



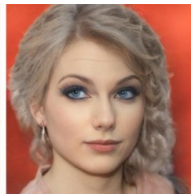
Input



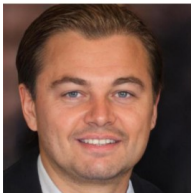
“Beyonce”



“A woman
without
makeup”



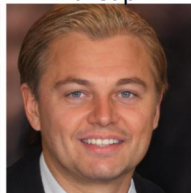
“Elsa from Frozen”



Input



“A man with
...”



“A blonde man”



“Donald Trump”



Input



Trees



Clouds



Spires



Round Roof



Advanced Diffusion Models: **Limitations**

Challenge	Explanation
Compute cost	Denoising many steps can be slow / costly
Biases & ethics	Models reflect biased training data
Guide artifacts	High guidance can oversharpen or distort
Fine-tune data needs	Requires quality, diverse data
Legal concerns	Generated content may face IP issues

Advanced Diffusion Models: **References**

- [1] Rombach, Robin, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. "High-resolution image synthesis with latent diffusion models." In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10684-10695. 2022.
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- [6] Ramesh, Aditya, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. "Hierarchical text-conditional image generation with clip latents." In *arXiv preprint arXiv:2204.06125*, 2022.
- [7] Abbeel, Pieter, Wilson Yan, Kevin Frans, and Philipp Wu. "CS294-158 Deep Unsupervised Learning, Lecture 6: Diffusion Models." UC Berkeley, 2022.
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